

WASP-49b

R-1337

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-49_141104 · created 2026-07-31T21:12:41+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456965.8993 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.112 +/- 0.012
Transit depth (Rp/Rs)^2	1.26 +/- 0.26 [%]
Orbital Inclination (inc)	84.47 +/- 0.44
Airmass coefficient 1 (a1)	1.203 +/- 0.0099
Airmass coefficient 2 (a2)	-0.1131 +/- 0.0045
Scatter in the residuals of the lightcurve fit is	0.82 %
Best Comparison Star	None
Optimal Aperture	4.08
Optimal Annulus	11.72
Transit Duration (day)	0.0883 +/- 0.0064

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.112 ± 0.012 vs 0.116 (deviation 0.33σ)
Duration fit vs published	0.0883 d vs 0.09 d (deviation 0.27σ)
Mid-time O–C	-0.9 min
Depth SNR	9.3
Residual scatter	0.82 %

Light curve

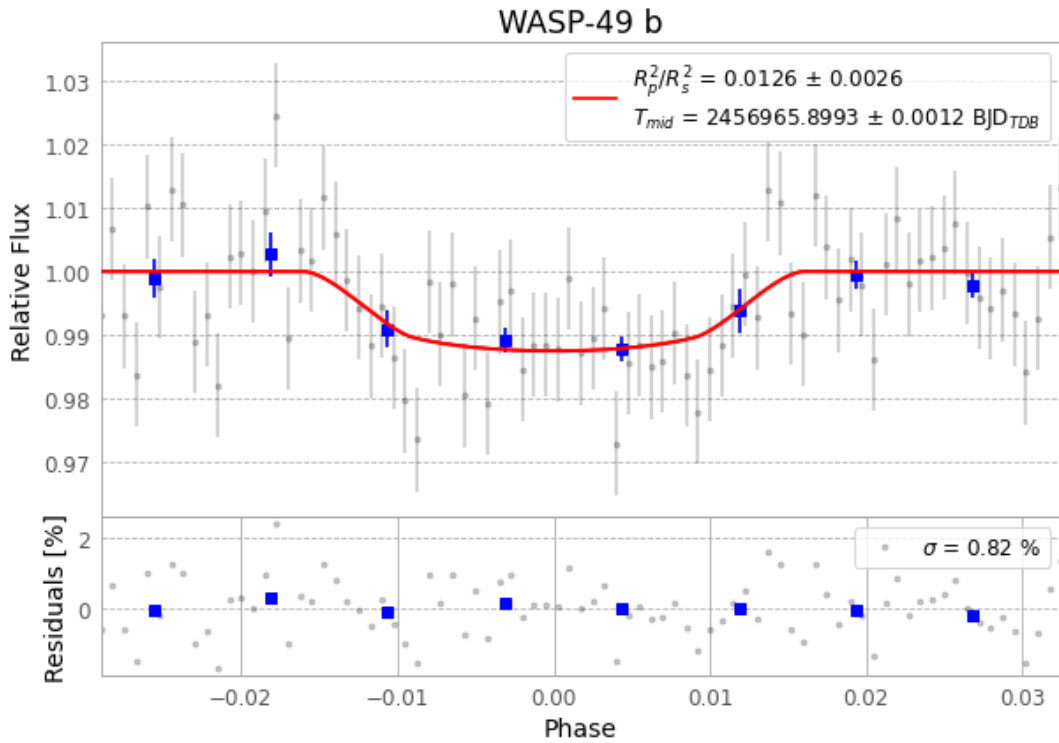


Figure 1 — FinalLightCurve_WASP-49 b_2014-11-04.png (SHA-256 2f09f93b5f3e3814...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [376, 212], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 bb8f51cd796493a3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

83 FITS frames (55 MB), WASP-49141104073312.FITS → WASP-49141104113912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-49-141104.log (00408e0db294...), FinalLightCurve_WASP-49 b_2014-11-04.png (2f09f93b5f3e...), FinalLightCurve_WASP-49 b_2014-11-04.csv (031a1c63015e...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 21:12Z.